Evaluation of the Validation Matrix

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Department of Technology,
Management and Economics

What were the results of your test?	What were your actions as a result of the test?	What is the assumption now? (Validated, a new, or the same?)	What would you do next if you had more time?
We created a mathematical model and concluded that 75% of the roof area produces enough energy	As our assumption was correct, so we didn't take any action and haven't had to adapt our prototype	It's the same, we didn't change it	We choose a simple mathematical mode assuming a fixed roof area and a fixed efficiency value per house. If we had more time, we would make the model more complex and try to include detailed roof top area and solar panel efficiency estimations.
Our survey showed that approximately 10% of Bornholm residents wouldn't like to participate	No action were needed so we continued with our prototype	It's the same, we didn't change it	Due to the limited time, we only got feedback from 15 Bornholm residents, with more time we could ask more residents
For the prototype setup we chose the energy and heat production is right in the spa that we calculated it would need to be	We didn't need to take any actions	It's the same, we didn't change it	We don't know which setup BEOF is preferring and we haven't had the time to talk with them. With more time we would have ask if our setup matches their needs
Our mathematical models showed that our setup provides enough energy during the spring, summer and fall. Only winters cold be challenging	We had to adjust our solution to be functional for sure during spring, summer and fall. For winter we can't guaranty 100% uptime and some biomass burning might be needed	We limit our assumption be true only during spring, summer and fall	Our model is quite simple using the average hour of sun for every season and the same efficiency value per house. If we had more time, we would have made the model more complex by including the directional change of the sun for every season and make the efficiency value for every house more accurate.

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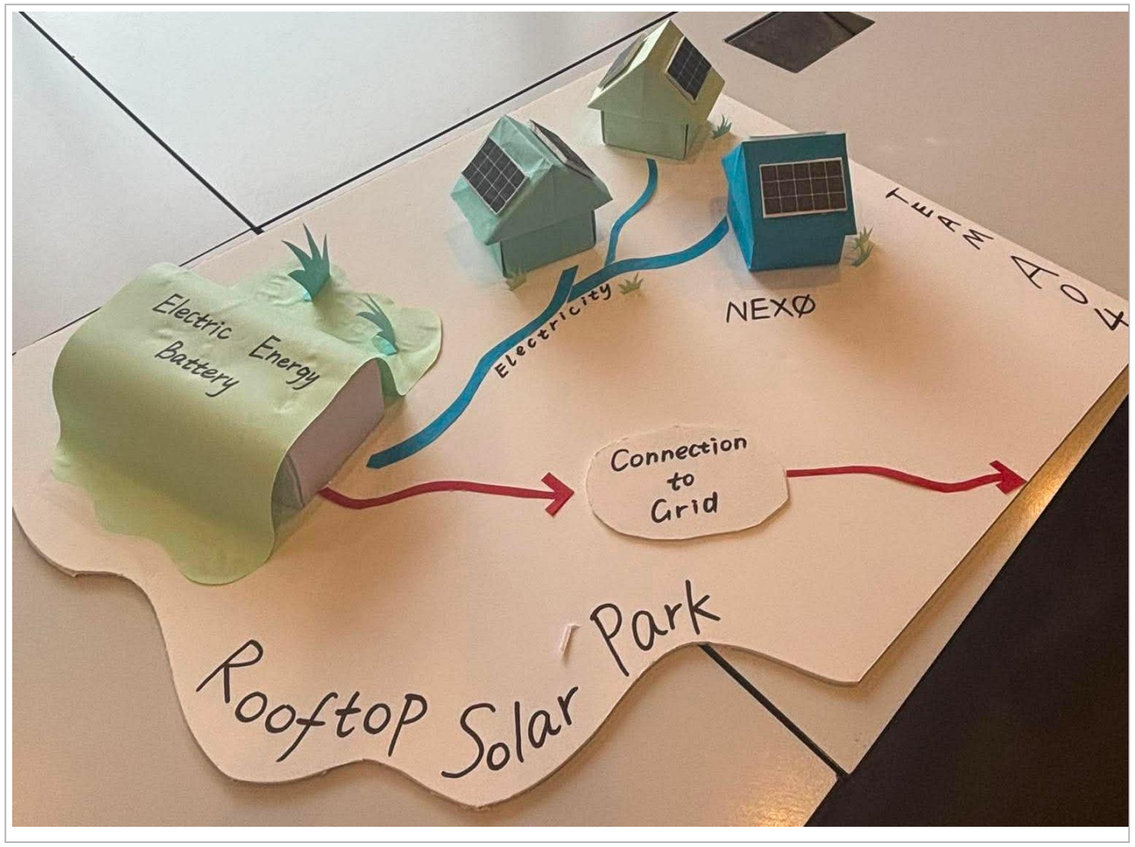
Present your prototype(s)

Title: Low-Fidelity Prototype of The Nexø Roof Top Solar Park

Introduce your prototype:

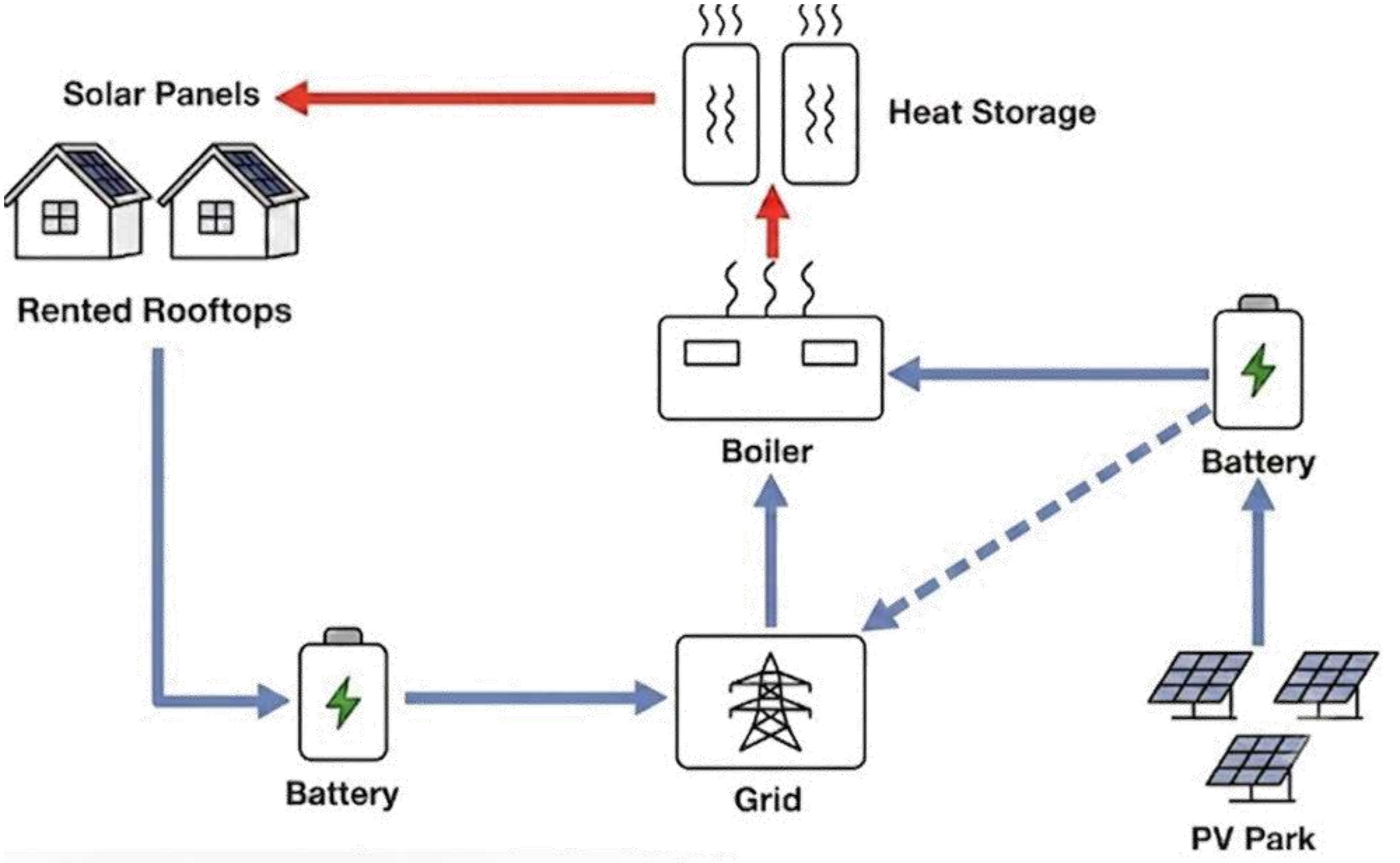
Our prototype is a low-fidelity visual model of The Nexø Roof Top PV. It shows how rooftop photovoltaic (PV), a PV park, battery storage, an electric boiler, heat storage, households and the electricity grid could be connected.
The prototype visualizes the flow of electricity and heat between the different components and shows how renewable energy and storage could work together to reduce pressure on the local electricity grid.
The prototype was used to test whether the concept is understandable to residents and to sell the idea to them, including the proposed energy flow is understandable , whether the combination of technologies makes sense, and whether any important connections or components are missing.
Based on the test and feedback, the overall system structure was understandable. However, the feedback also highlighted that the solution depends on residents' willingness to participate, which needs to be considered further.

Insert pictures of your prototype



If you have any links for videos, mockups, etc

Model


Solar Panels
Heat Storage
Rented Rooftops
Boiler
Battery
Battery
Grid
PV Park